



INTEGRATING WAVELET SCATTERING INTO CNN

Deep Learning 00460217 Technion

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https://github.com/baraktelem/deep_learning_course_project

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Introduction

Project Goal

The primary objective of this project is to explore the integration of analytic wavelet priors - specifically Gabor filters – into the architecture of CNNs. While standard CNNs learn filters from scratch, often resulting in redundant or noisy feature extractors in the early layers, we propose novel “Hybrid” architectures that inject strong inductive biases directly into the network structure.

Specifically, we aim to:

1. **Develop Hybrid Architectures:** Design and implement neural network layers that combine standard learnable convolutions with fixed and parameterized Gabor filters.
2. **Enhance Sample Efficiency in Hybrid Models:** Systematically quantify the performance of our proposed hybrid architectures in low-data regimes. We aim to determine if our specific method of partial constraint integration yields the expected theoretical efficiency gains compared to fully learnable baselines.
3. **Mitigate Texture Bias:** Examine if explicit frequency-based filtering can shift the network’s focus from high-frequency textures to more robust shape-based features, potentially improving generalization.
4. **Optimize Resource Utilization:** Analyze whether these hybrid layers can reduce the number of learnable parameters without sacrificing accuracy, thereby acting as a form of structural regularization.

Motivation

CNNs have revolutionized computer vision, yet their training process remains computationally expensive and data-hungry. Our motivation stems from the observation that the early layers of converged CNNs consistently converge to filters resembling Gabor wavelets and color blobs, regardless of the dataset. This suggests a redundancy in the standard training paradigm: the network spends significant resources “re-learning” known edge-detection operators from scratch.

By integrating predetermined, biologically plausible filters into the architecture, we aim to address three key challenges:

1. **Sample Efficiency via Inductive Bias:** Standard CNNs possess a weak inductive bias. By replacing a portion of the early filters with analytic Gabor kernels – mimicking the simple cells in the mammalian primary visual cortex – we introduce a strong structural prior. We hypothesize that this “hard-coded” knowledge allows the network to learn class-specific features with fewer examples, as it does not need to waste data discovering fundamental edge detectors.
2. **Mitigating Texture Bias:** Deep networks are known to rely heavily on high-frequency texture cues rather than global shape. By offloading texture detection to a dedicated, constrained wavelet pathway, we aim to “free up” the remaining standard convolutional filters to focus on higher-level semantic structures and shapes.
3. **Partial Structural Regularization:** Finally, we view the hybridization of fixed and learnable parameters as a form of flexible regularization. Unlike a pure wavelet network (which is rigid) or a standard CNN (which is prone to overfitting noise), our approach seeks a middle ground. We are motivated to explore if this mixed architecture – combining the robustness of fixed priors with the generalization capabilities of standard learning – improves resilience to common corruptions like noise and blur.

Background

- Wavelets and Gabor Filters: Deep learning models, particularly CNNs, often converge to filters that resemble Gabor wavelets in their initial layers. Gabor filters are band-pass filters that localize features in both space and frequency, mimicking the impulse response of simple cells in the mammalian primary visual cortex. Unlike learned kernels which can degenerate into noise, analytic Gabor functions provide optimal joint resolution in the spatial and frequency domains.
- Texture Bias in CNNs: Standard CNNs are known to be biased towards texture rather than shape. This “texture bias” makes them susceptible to distortions that preserve texture but destroy shape. Previous research suggests that increasing shape bias can improve model robustness.

Previous Work

Research into integrating fixed priors with deep learning generally falls into categories that influenced our work:

- Invariant Scattering Networks: Bruna and Mallat proposed the Scattering Transform, which cascades fixed wavelet convolutions and modulus non-linearities to create translation-invariant representations. While stable, these networks traditionally lack the adaptability of supervised learning.
- Parametric Scattering Networks: To bridge the gap between fixed priors and learning, Gauthier et al. introduced Parametric Scattering Networks, where the parameters of the wavelets (scale, orientation) are learned during training.
- Our Distinction: While previous works often focus on replacing CNN layers entirely or using complex scattering manifolds, our project explores a hybrid approach. We integrate Gabor kernels (both fixed and learnable) alongside standard convolutions, aiming to offload texture detection while retaining the flexibility of the standard ResNet Backbone

Method

To investigate the impact of structural priors on sample efficiency and robustness, we developed a set of “Hybrid” layers that combine standard learnable convolutions with analytic frequency-based filters. We integrated these layers into a ResNet18 backbone, replacing standard convolutional blocks at various depths.

Hybrid Layer Designs

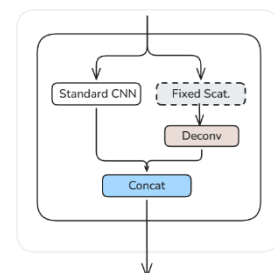
We explored two primary strategies for integrating these priors, differing in both learnability and integration method.

Fixed Scattering (Channel-wise Only)

Our initial approach utilized the Wavelet Scattering Transform to inject fixed, translation-invariant features.

Mechanism: The layer splits the channel budget.

1. Feature Generation: We generate fixed texture descriptors using a Scattering Transform.
2. Resolution Matching: Since the scattering transform downsamples the input, we apply a Transposed Convolution layer with stride 2 and kernel size 2 to upsample the features back to the required output spatial resolution.
3. Concatenation: These upsampled maps are concatenated with the standard learned features.



Parametric Hybrid (Channel-wise & Element-wise)

While the fixed scattering transform offers strong mathematical guarantees – such as local translation invariance and stability to deformations - it imposes a rigid, “one-size-fits-all” filter bank. This generic prior may be suboptimal for specific tasks where discriminative features reside in frequency bands or orientations not perfectly covered by the fixed dyadic grid.

To bridge the gap, we developed a Parametric Hybrid Layer. This approach retains the inductive bias of the Gabor structure but relaxed the rigidity of the parameters. By making the Gabor parameters learnable, we allow the network to tune its biological priors to the specific statistics of the dataset, moving from a static “hard-coded” prior to a data-adaptive one.

Mechanism and Optimization

Unlike standard convolutional layers that store a static weight matrix, our parametric layer learns the generative parameters of the Gabor function.

- **Parameter Selection:** The full Gabor formulation includes parameters for phase and aspect ratio. However, based on brief initial testing, we decided to focus optimization on the three primary geometric parameters: orientation, scale, and wavelength.
- **On-the-Fly Generation:** During each forward pass, the $K \times K$ pixel-wise Gabor filters are generated physically from these three learnable parameters using the formulation above.
- **Optimization:** We utilize standard gradient descent allowing gradients to flow through this generation step via the chain rule. Crucially, we treat these geometric parameters differently from standard weights: we disable weight decay and apply a higher learning rate as empirical tuning suggested this yields improved performance.
- **Initialization:** To ensure diverse feature extraction, we initialize the parameters to span a uniform distribution of orientations and frequencies.

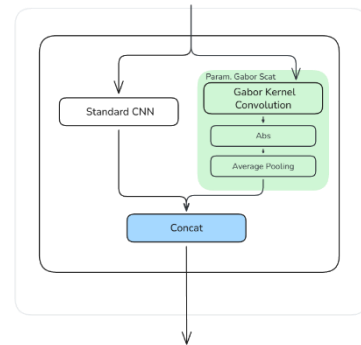
Integration Strategies

We investigated two distinct fusion strategies. In both cases, we designed the layers to act as drop-in replacements for standard convolutions, maintaining the backbone’s feature map dimensions

Channel-wise Integration

We designed this variant to emulate the Scattering Transform’s stability properties (modulus nonlinearity + average pooling) within a learnable framework. Unlike the fixed transform, we utilize stride-1 pooling to preserve spatial resolution. Crucially, we split the total channel budget between the standard and Gabor paths.

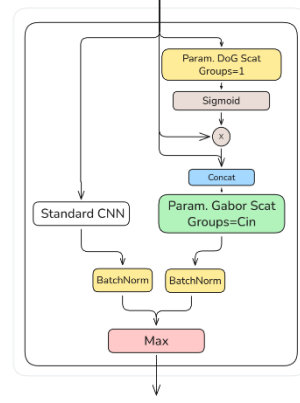
This allows us to replace layers without increasing the network width, effectively substituting a portion of the purely learnable capacity with stable, structural priors.



Element-wise Integration (Competitive Max-Out)

This stricter approach maintains the full channel count for both paths and forces them to compete.

- DoG Attention: The Gabor path includes a Difference of Gaussians (DoG) attention gate (same scattering mechanism but with DoG kernels and groups=1). While intended to serve as a spatial attention mechanism, we note that we did not isolate this component’s specific impact in our experiments.
- Fusion: The final output is derived via an element-wise maximum. This competitive mechanism allows the network to dynamically “select” the biological prior only when it produces a stronger response than the standard learned filter.



Architectures

We utilized ResNet18 as our backbone. We established a baseline by training the standard ResNet18 from scratch and then systematically introduced our hybrid layers to evaluate their impact.

Layer 1 Replacement

The first convolutional layer (Conv1) of a ResNet is responsible for extracting initial low-level features. We focused our primary comparative analysis here, replacing this single layer with our three variants while keeping the rest of the network standard:

- ScatResNet: Replaces Conv1 with the Fixed Scattering Hybrid (Concatenation).
- HybridResNet: Replaces Conv1 with the Parametric Hybrid (Concatenation).
- MaxHybridResNet: Replaces Conv1 with the Parametric Hybrid (Element-wise maximum)

Deep Integration (Progressive Replacement)

Based on the performance of the initial L1 experiments, we selected the Max-Hybrid architecture for deeper integration testing. We defined a progression of models $L = 1$ to $L = 5$ where L represents the number of consecutive convolutional layers replaced by our hybrid layer. This allows us to determine if Gabor priors remain beneficial as feature abstraction increases or if their utility is confined to the initial pixel-level processing.

Experiments and Results

Experimental Setup

We trained ResNet18-based models on the CIFAR-10 dataset for 200 epochs with a batch size of 128. Optimization was performed using SGD (momentum 0.9, weight decay $5e^{-4}$, initial learning rate 0.1) coupled with a Cosine Annealing scheduler ($T_{max} = 200$). To evaluate sample efficiency, models were trained on stratified subsets of the training data, validated on 5000 samples and evaluated on the full test set.

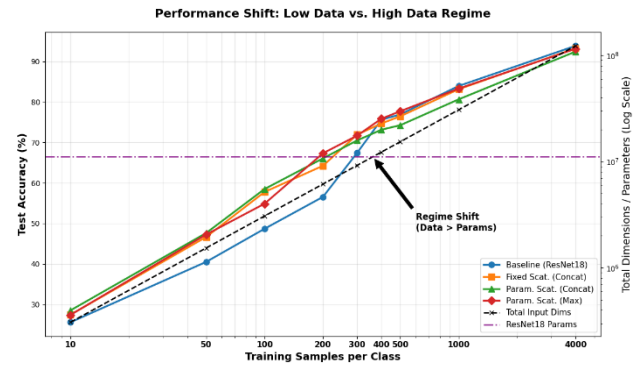
First Layer Experiment

We evaluated the impact of replacing the initial convolutional layer (Conv1) with our three hybrid variants: fixed scattering concatenation, parametric scattering concatenation and parametric scattering max-out. We compared their sample efficiency against the standard ResNet18 baseline across varying dataset sizes.

Results: Sample Efficiency & Regimes

We observed distinct performance behaviors across three data regimes:

- **Low-Data Regime (< 500 samples per class):** All hybrid architectures significantly outperformed the baseline. This performance gap can be attributed to overfitting prevention. The baseline model achieves high training accuracy but poor test accuracy in this regime, indicating rapid overfitting. The hybrid layers, by imposing structural priors, restrict the solution space and act as a strong architectural regularizer, preventing the model from memorizing noise.
- **Middle Regime (200-500 samples per class):** The hybrid variants performed similarly, though the parametric scattering max-out displayed a slight advantage over the concatenation-based hybrid variants. This suggests that the competitive max-out mechanism may offer a more robust feature selection strategy when data is sufficient to learn basic patterns but insufficient to refine them into robust features detectors, the max operation allows the network to automatically ‘fall back’ on the stable Gabor features whenever the learned filters are weak or noisy.
- **High-Data Regime (≥ 500 samples per class):** As data availability increased, the baseline surpassed the hybrid models, though the gap remained minimal for the max-out and fixed variants. Notably, the parametric concatenation model underperformed the fixed scattering baseline in this regime. This likely reflects a trade-off: while the fixed transform guarantees signal preservation, learning the parameters can disrupt these properties. We attribute this drop to the difficulty of optimizing the geometric parameters and potential limitations in our generator design, which may have prevented the learnable filters from converging to effective feature extractors.



Deep Integration Experiment

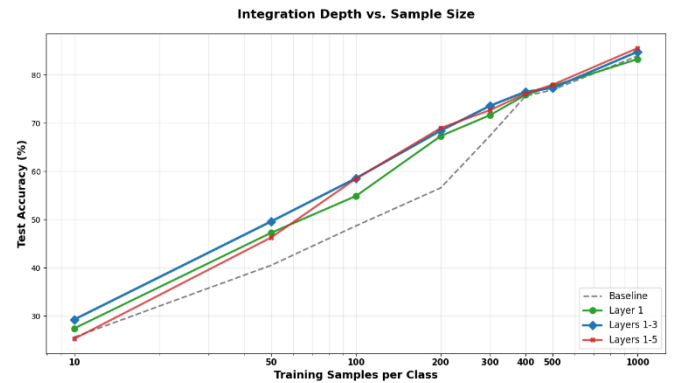
We extended the analysis by integrating hybrid layers deeper into the network, specifically focusing on the max-out variant.

Rationale: We selected the max-out variant for this experiment because its competitive nature acts as a gating mechanism—preserving the full capacity of the standard convolutional path while allowing the Gabor priors to override learned features when the biological signal is stronger. Unlike the concatenation strategy, this approach strictly augments the standard features without reducing the learnable parameter space in these critical early layers. While the channel-replacement strategy (Concatenation) offers parameter reduction, we reasoned this was less critical in the early layers, as the vast majority of ResNet parameters reside in the deeper stages.

Results: Impact of Depth on Sample Efficiency

Comparing the sample efficiency curves for varying replacement depths revealed the following trends:

- Diminishing Returns: While adding priors to deeper layers improved performance over the baseline, the marginal gain decreased compared to the initial impact of replacing the first layer.
- L3 vs. L1: Replacing the stem and the first full residual block (L1-3) performed slightly better than the single layer replacement (L=1) across almost all data regimes, with the exception of the 500-samples per class case where performance was comparable.
- Deepest Integration: The deepest replacement tested (L=1-5) showed a slight improvement over the baseline in the highest data regime tested (1000 samples per class), suggesting that deep structural priors may aid generalization when moderate data is available.



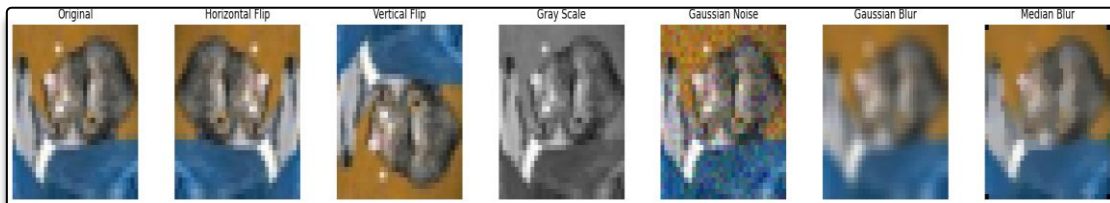
Computational Constraints: We restricted this analysis to sample sizes ≤ 1000 per class. Extending the hybrid layers to deeper blocks – which contains significantly more channels increased the computational cost of the on-the-fly filter generation. Training time for the deep hybrid model L1-5 was approximately 5 times slower than the baseline, making evaluation on larger datasets prohibitive for this report.

Robustness Experiment

Experimental Setup

To evaluate the generalization capabilities of our models, we assessed their robustness against two primary categories of variations: textual variations (as highlighted in [2]) and geometric transformations (as seen in class). We utilized the Kornia library to apply a suite of six distinct augmentations on the test set. These were organized into three subgroups to isolate specific invariance properties:

1. Positional Changes: Horizontal and Vertical Flips to test geometric invariance.
2. Texture Perturbations: Comprising both Color Changes (Grayscale, Gaussian Noise) and Blur operations (Gaussian Blur, Median Blur) to challenge the model's reliance on high-frequency texture cues.



We tested both classification accuracy and accuracy drop (robustness) on 4 models:

1. ResNet18 – Baseline
2. ScatResNet18 – ResNet18 with fixed scattering hybrid layer.
3. MaxHybridGaborLayer2 – ResNet18 with 2 first layers replaced with learnable scattering
4. MaxHybridGaborLayer3 – ResNet18 with 3 first layers replaced with learnable scattering

For further comparison, we tested those metrics as functions of data size.

Results and Analysis

The classification accuracy results can be seen here:

We can see that in the low data regimes, in most cases, scattering (fixed or learned) still performs better than ResNet.

The exception is in the case of Gaussian noise where fixed scattering performs much worse but learnable scattering performs better in all data sizes. Moreover, fixed scattering performs better on median blur in all data sizes.

The accuracy drop (robustness) can be seen here:

Regarding positional augmentations, all evaluated models exhibit a comparable magnitude of accuracy degradation.

Given the significant noise observed in the performance curves, we focus our further analysis on comparing fixed scattering with ResNet in the more stable high-data regime.

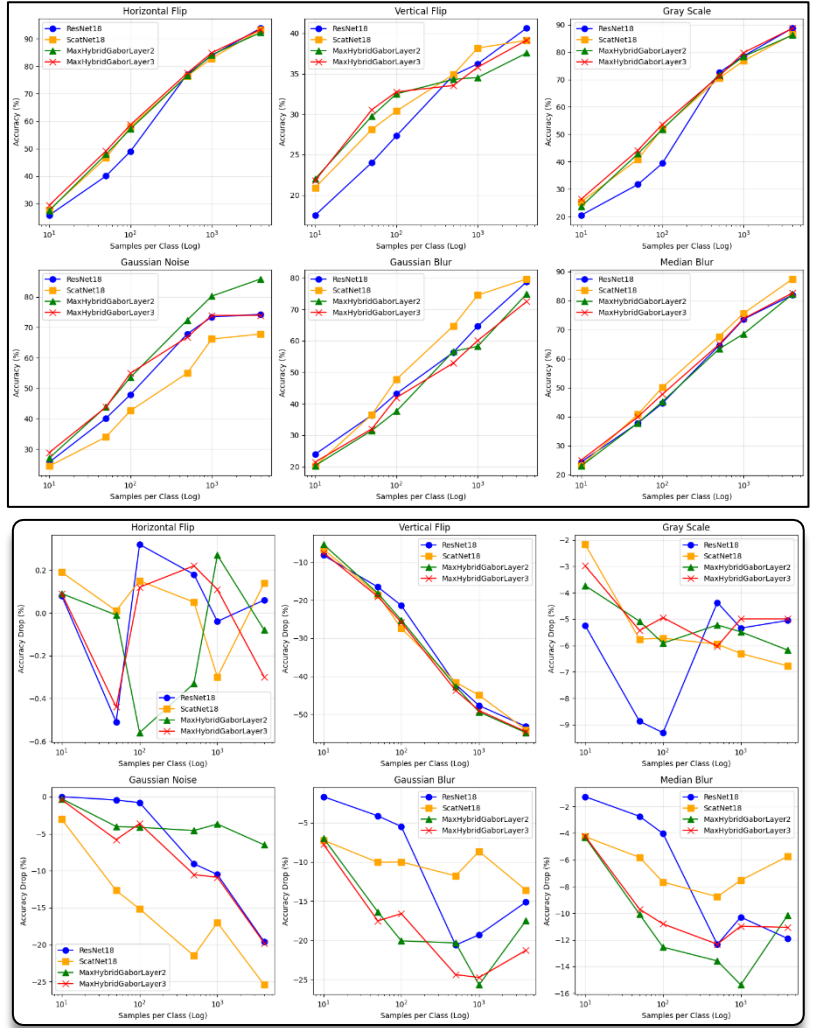
We observe distinct performance divergences: while fixed scattering is significantly more sensitive to Grayscale and Gaussian Noise than ResNet, it considerably outperforms the baseline on Gaussian and Median Blur. We hypothesize this stems from the scattering transform's inherent focus on high-frequency texture details. Noise acts as a high-frequency corruption, triggering spurious high activations in the scattering channels that disrupt classification.

In contrast, blur operations function as low-pass filters. This suppression of high frequencies likely forces the classification head to bypass the inactive scattering coefficients and rely on the learnable CNN kernels, which process the remaining low-frequency information effectively. This hypothesis is supported by the performance gap between Median and Gaussian Blur; the drop in accuracy is notably lower for Median Blur (6-8%) compared to Gaussian Blur (8-14%). Since Median Blur preserves edge structures (valid high-frequency components) better than Gaussian Blur, the scattering transform is able to leverage this retained information, resulting in higher robustness.

These findings validate our initial theory: by delegating texture detection to the scattering layers, the CNN kernels are liberated to learn complementary structural features.

Conclusion

We demonstrated that integrating Gabor priors into CNNs significantly improves sample efficiency in low-data regimes (<500 samples per class) by acting as a structural regularizer against overfitting. Our experiments highlighted three key findings:



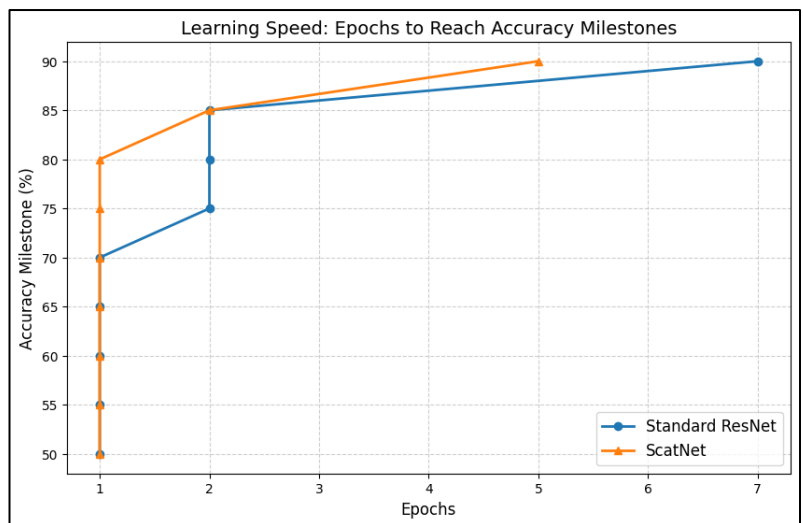
- Sample Efficiency: Hybrid models consistently outperformed the baseline when data was scarce.
- Depth Dependence: Structural priors proved most effective in the early network stage ($L = 1$ to 3). Extending them deeper ($L=5$) yielded diminishing returns, confirming that analytic priors are best suited for early visual processing rather than deep semantic abstraction.
- Robustness Trade-off: A distinct trade-off emerged: hybrid models showed superior robustness to blur but increased sensitivity to high frequency noise. This suggests that offloading texture detection to wavelet pathways encourages the learned kernels to focus on more robust, low-frequency structural features.

Future Work

Transfer Learning

The observed performance gains in the low-data regime suggest that scattering integration enhances the network's adaptability. To empirically validate this, we conducted a transfer learning experiment designed as a preliminary proof of concept: both models were fully pre-trained on an initial 5-class subset and then fine-tuned on a distinct set of 5 classes. We quantified adaptation speed by tracking the number of epochs required to reach specific accuracy milestones on the new task.

As illustrated in the graph, the fixed scattering model (ScatNet) converges faster to the new data, reaching 90% accuracy in fewer epochs than the baseline.



Texture Bias

Our robustness experiment suggests that while scattering transforms may effectively encode texture, they may inadvertently induce a strong texture bias in the CNN. This over-reliance is evidenced by the disproportionate drop in accuracy when high-frequency information is corrupted (e.g., Gaussian Noise, Grayscale). Future work could investigate specific regularization techniques applied to the scattering channels. By penalizing over-activation in these layers, we aim to alleviate this "texture fixation," encouraging the network to learn a more balanced representation that integrates both textural and structural features.

References

1. [Parametric Scattering Networks \(Gauthier et al.\)](#)
2. [ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness \(Geirhos et al.\)](#)
3. [Invariant Scattering Convolution Networks \(Bruna et al.\)](#)
4. [Kymatio: Scattering Transforms in Python \(Andreux et al.\)](#)

Ethics Statement

1. Introduction

Student names: Gilad Navok, Barak Telem

Project Title: Integrating Wavelet Scattering into CNN

Description: In our project we to examine integrating wavelet scattering transform (WST) with CNNs. We wanted to see if different types of integrations can improve various performance metrics.

2. Questions for LLM (Gemini)

2a. List 3 types of stakeholders that will be affected by the project.

1. Deep Learning Researchers: Individuals in academia or R&D interested in improving CNN architectures, specifically regarding texture bias and geometric priors like Wavelet Scattering.
2. Industry Practitioners in Low-Data Regimes: Data scientists and engineers working in fields where labeled data is scarce (e.g., medical imaging or specialized manufacturing), who would benefit from the "Small data training" capabilities and improved sample efficiency.
3. Project Supervisors & Academic Evaluators: The faculty members or teaching staff responsible for overseeing the project's completion, grading the "Results", and validating the "Motivation" and methodology.

2b. What will an explanation that is given to each stakeholder look like?

For Researchers, the explanation will focus on the theoretical integration of fixed and parametric scattering priors into ResNet18, highlighting how this hybrid approach effectively encodes texture but introduces specific robustness trade-offs (e.g., sensitivity to noise vs. invariance to blur). For Industry Practitioners, the explanation will emphasize the practical utility: the ScatNet model demonstrates significantly faster convergence speeds and superior accuracy in low-data regimes (10-100 samples per class), making it a cost-effective solution for limited datasets. For Project Supervisors, the explanation will present a comprehensive summary of the experimental design, confirming that the hypothesis was validated- specifically, that replacing early CNN layers with scattering transforms accelerates training and improves data efficiency, while also conducting a rigorous ablation study on integration depth.

2c. Who is responsible for giving the explanation to each stakeholder?

The primary responsibility for communicating these findings lies with the project authors, Gilad Navok and Barak Telem. They are responsible for delivering the academic presentation and final report to their supervisors to demonstrate project success. Additionally, should this work be published or open-sourced, Gilad and Barak would be responsible for creating the technical documentation and paper abstract required to explain the methodology and results to Researchers and Industry Practitioners.

3. Reflection on AI Output

We think that the main change in Gemini response to be more ethical is emphasizing that we worked on our project with very few resources (we used Kaggle for all our training and experiments). Due to this constraint, we decided to work on images resolution of 32X32. This low resolution may not be representative for higher resolution images. Furthermore, it should be rigorously tested and confirmed before applying to the fields Gemini suggested (e.g., medical imaging, specialized manufacturing).